MOHAMMAD KHALILUR RAHMAN

MAINTENANCE INSTRUMENTATION & AUTOMATION ENGINEER

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POSITION APPLIED FOR

Maintenance Instrumentation and Automation Engineer — Rebar and Coil Hot & Cold Rolling Mill / Steel Plant, including steelmaking, sand casting and open-die forging.

CAREER OBJECTIVE

To be part of a growth-oriented and challenging environment where technical, analytical and managerial skills can be utilized effectively, contributing to the growth of the company as well as professional development.

PROFESSIONAL SUMMARY

Instrumentation and Automation Engineer with 18+ years of stated total work experience, including 16+ years of Gulf experience as recorded in the original resume. Experience covers maintenance, instrumentation, automation and troubleshooting in steel plants, rebar and coil rolling mills, steelmaking, sand casting and open-die forging. Technical exposure includes PLCs, HMI/SCADA, AC/DC drives, process instruments, calibration, electrical diagrams, MV/LV distribution equipment, shutdown planning, spare-parts coordination and manpower supervision.

WORK EXPERIENCE

Total Work Experience: 18+ years (as stated in the original resume)

Gulf Experience: 16+ years (as stated in the original resume)

Maintenance Instrument and Automation Supervisor | Tuwaiq Casting & Forging

Ras Al Khair, Saudi Arabia • 25 August 2026 – Present

- Working in maintenance instrumentation and automation for plant operations covering steelmaking, sand casting and open-die forging.
- Plant technology exposure includes Siemens S7-1500 and ET 200SP over PROFINET, ABB and Siemens protection relays, ABB drives, and WinCC operator and process workstations.
- Plant scope includes steelmaking, sand casting, open-die forging and machining.

Maintenance Instrument and Automation – Senior Technical Role | SOLB Steel Company (SOLB)

Jizan Economic City, Saudi Arabia • 18 February 2013 – 25 August 2026

- Maintenance and automation work in a rebar and coil rolling mill steel plant.
- Conducted fault and trip analysis, troubleshooting and technical problem-solving for plant instrumentation, control and automation equipment.
- Worked with PLCs, HMI/SCADA, ABB AC/DC drives, sensors, encoders, process-control instruments and related equipment used in rolling mill operations.
- Performed instrument calibration and supported preventive maintenance, breakdown rectification, shutdown work and electrical maintenance.
- Supported maintenance planning, spare-parts coordination, work prioritization and technician task allocation.

Automation Engineer (Rolling Mill) | AL-YAMAMAH STEEL

Yanbu, Saudi Arabia • 01 October 2010 – 18 January 2013

- Worked as an Automation Engineer supporting rolling mill automation systems and associated control equipment.
- Supported maintenance and troubleshooting activities related to automation and rolling mill operation.

Maintenance Instrument and Automation Engineer (Rolling Mill) | Tata Tiscon, Tata Iron & Steel Co. Ltd.

Jamshedpur, India • 14 September 2004 – 10 July 2006

- Worked in maintenance instrumentation and automation for rolling mill operations.
- Supported equipment maintenance, fault identification and troubleshooting in the rolling mill environment.

FAMILIARIZATION WITH ADVANCED TECHNOLOGY IN THE FIELD

- Plant ownership background: a joint venture of Dussur, Saudi Aramco and Doosan, with Doosan as the technical partner.
- Plant scope: integrated plant covering steelmaking, sand casting, open-die forging and machining.
- Steelmaking automation: Danieli DIGIMET automation for the EAF (Electric Arc Furnace), LRF (Ladle Refining Furnace) and VOD, with WinCC operator and process workstations.
- Forging equipment: SMS group open-die presses, including a 40/45 MN four-column press and a 20/22.5 MN two-column press.
- Control and electrical systems: Siemens S7-1500 and ET 200SP over PROFINET, ABB and Siemens protection relays, and ABB drives.
- Issues identified in the original notes: shell-cooling RTD faults (B001 and B081), trips associated with Water Cooling and Common Parts, and network faults affecting some devices.
- Fully automatic, high-speed rebar rolling mill design by Danieli.
- Fully automatic rebar rolling mill design by SMS MEER.
- Cut-and-bend steel forming and coating operations (Steefco): straight plain and deformed rebar in sizes from 5 mm to 10 mm; cut-and-bend steel in various shapes and diameters from 8 mm to 40 mm; welded wire mesh; and FBEC epoxy-coated steel rebar from 8 mm to 40 mm.
- Multi-level automation systems used in the steel industry.
- SIMATIC industrial automation systems.
- ABB drives, including ACS800, DCS800, MCCD and soft starters.
- Siemens drive systems, including SINAMICS Power Module 340 and AC-to-AC converters.
- SIMATIC HMI WinCC, Wonderware InTouch V6 and OWS (Operator Work Station).
- Schneider Altivar 312 variable-frequency drive (VFD).
- Reading and understanding electrical diagrams for rebar rolling mills and heavy-industry systems.
- IBA AG analyser.
- TCP/IP computer networking systems.
- ABB medium-voltage switchgear and 33 kV power-distribution switchboard.
- Ten step-down three-phase transformers, including one spare transformer for breakdown situations.
- Low-voltage 380 V, three-phase distribution switchboards and ABB power-centre switchboards PC01, PC02, PC03 and PC04.
- Emergency power system: 380 V, 60 Hz diesel generator supplied by VISA.
- ABB ACS800 multi-drive system with 33 kV input from a transformer.
- Eaton UPS systems: UPS.01 at 110 V DC and UPS.02 at 220 V AC.
- High-bay lighting and lighting-distribution systems, including MLDB, ELDB and SLDB.
- Calibration of process measurement and control instruments, including flow meters, level sensors, distance sensors, loop scanners, hot-metal detectors, thermocouples, flow indicators, resistance temperature detectors and pyrometers.
- CCTV systems using IP networking and analogue cameras.

SOFTWARE EXPOSURE

- Operating systems: Windows 2000, Windows Server 2003, Windows XP and Windows Vista.
- Office packages: Microsoft Office 2000, 2003, 2007 and 2011, and Microsoft Access.
- DCS and SCADA: SIMATIC WinCC.
- PLCs: Siemens STEP 7 SIMATIC Manager and ABB Control Builder M.
- HMIs: Siemens WinCC.
- Drive software: ABB DriveWindow and Siemens DriveWindow.

JOB RESPONSIBILITIES

Maintenance During Operations

- Conduct fault and trip analysis, troubleshoot equipment problems, and advise on suitable technical solutions. Carry out regular inspections to help prevent accidents and comply with safety requirements.
- Carry out installation, commissioning, repair, maintenance and troubleshooting of electronic devices and sensors used in rolling mills, including photodiodes, proximity switches, position switches, thermistors and resistance temperature detectors.
- Calibrate instruments used for measurement and control, including flow meters, level sensors, distance sensors, loop scanners, hot-metal detectors, thermocouples, flow indicators, resistance temperature detectors and pyrometers.

- Work with field control elements such as thermocouples, PT100 RTDs, gauges, switches, loop scanners, hot-metal detectors, pyrometers, recorders, PID converters, controllers and control valves, including their associated parameters.
- Work with programmable logic controllers (PLCs), including ladder diagram (LD) and STL logic; read and check logic, bypass interlocks when required and authorized, add or remove logic, and work with analogue and digital PLC inputs and outputs.
- Troubleshoot SIMATIC 400, 300 and 200 PLC control panels; identify faults involving major components and understand their functions.
- Work with Siemens digital control systems, S5 and S7 PLC systems, PLC converters, AC/DC drives, controllers and HMIs.
- Work with SMS AG technological control systems, networking, DAS and measuring systems.
- Administer Level-2 servers and mill setup computers and related server systems.
- Understand and support control-room, furnace and mill operations, including main pulpit operations, CD1, finishing areas, combined cut-length (CCL) equipment, tying-machine areas, CD2 operations and troubleshooting.
- Troubleshoot HMI (Human-Machine Interface) issues and change parameters according to operational requirements, including OWS (Operator Work Station) data when problems occur.
- Work with ABB ACS800 multi-drive systems supplied from a 33 kV transformer input.
- Maintain and troubleshoot ABB ACS800 AC motor drives and DCS800 DC motor drives, including changing parameters according to requirements.
- Maintain ABB ACS800 AC drives, check vibration, troubleshoot faults, perform load and no-load tests, and carry out calibration checks on DC and AC motors.
- Check encoders used on high-speed Danieli-designed bar rolling mills.
- Identify obstructions and equipment breakdowns, apply troubleshooting techniques to restore equipment, and implement cost-saving and energy-saving measures aimed at reducing operating costs.
- Develop and implement maintenance plans and schedules, including annual turnaround activities, with attention to effective use of manpower and materials.
- Maintain and troubleshoot circuit breakers, motors, transformers and different types of transducers.
- Reassemble repaired electric motors to specified requirements and ratings using hand tools and electrical meters.
- Maintain medium-voltage switchgear and low-voltage panels during annual shutdowns.
- Rewire electrical systems and repair or replace electrical accessories.
- Inspect electrical connections and read and follow electrical diagrams.
- Inspect and test equipment to locate damage or worn parts and diagnose malfunctions; use work orders and schematic drawings to determine required repairs.
- Troubleshoot and rectify problems during equipment start-up and shutdown, including critical conditions.
- Use electrical testing equipment, including a megger for insulation testing.
- Tighten, solder, wrap, flux and coat wires to support proper insulation and grounding.
- Calibrate process instruments in accordance with IMS (International Maintenance Standards), as stated in the original resume.

Shutdown Management

- Plan and support shutdowns that occur weekly or fortnightly, as required, with shutdown periods of at least 8 to 12 hours.
- Plan shutdown work at least four days in advance.
- Complete pending and backlog maintenance work during the scheduled shutdown period.
- Prepare a prioritized list of work before attending each shutdown.

Material Management

- Maintain rolling mill spare parts so that equipment work is not delayed because of missing spares.
- Raise purchase requisitions for required spare parts.
- Check incoming materials when received.
- Follow up on the status of ordered spare parts.
- Restore and maintain standby equipment as early as possible.

Manpower Management

- Assign tasks to team members and advise them on how to complete work safely and efficiently.
- Collect feedback from team members periodically.
- Prepare standard operating procedures (SOPs) for different types of work.
- Report maintenance activities and work progress to the head of department at regular intervals.

TECHNICAL / PROFESSIONAL QUALIFICATIONS

- Degree in Electronics & Communication Engineering — Shankara Institute of Technology, Jaipur; University of Rajasthan, Jaipur. Marks: 65%.
- Three-Year Diploma in Electronics & Communication Engineering — Al-Kabir Polytechnic, Jamshedpur; S.B.T.E., Ranchi. Marks: 74.04%.

ACADEMIC QUALIFICATIONS

- Intermediate (12th) — J.I.E.C., Ranchi; year of passing: 2001; marks: 61.00%.
- Matriculation / S.S.E. (10th) — B.S.E.B., Patna; year of passing: 1999; marks: 59.00%.

ADDITIONAL QUALIFICATIONS

- Certificate in Programmable Logic Control (PLC), S7-300 Micro PLCs — A.K.P., Jamshedpur.
- Certificate in Industrial Automation and Control — I.D.T.R., Jamshedpur.

PERSONAL INFORMATION

Date of Birth: 02 October 1983
Nationality: Indian
Religion: Islam
Marital Status: Single
Languages Known: English, Hindi and Urdu

PASSPORT DETAILS

Passport Number: S9755224
Date of Issue: 08 October 2018
Date of Expiry: 07 October 2028
Place of Issue: Jeddah

DECLARATION

I hereby certify that the statements above, including any attachments submitted now or later, are accurate to the best of my knowledge.